

AGADIR AL MASSIRA INTL, Morocco

WMO: 602520

Lat: 30.325N

Lon: 9.413W

Elev: 76

StdP: 100.41

Time Zone: 1.00 (E01)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	5.0	6.1	-5.2	2.5	23.6	-3.0	3.0	21.5	10.5	18.9	9.1	18.9	2.4	60	0.387

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	11.5	38.9	20.0	35.1	19.2	32.2	19.0	22.5	30.6	21.8	29.3	21.4	28.4	5.3	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.2	15.0	24.1	19.8	14.7	23.8	19.2	14.1	23.3	66.3	30.5	64.2	29.8	62.5	28.4	27.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.3	8.0	7.1	2.6	44.3	1.4	3.5	1.7	46.8	0.8	48.8	0.1	50.8	-0.9	53.3		
			1.7	23.9	1.6	1.2	0.6	24.7	-0.3	25.4	-1.2	26.1	-2.4	26.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	19.9	14.5	15.7	18.0	18.9	20.4	22.5	24.1	24.7	23.2	22.1	18.6	15.7
	DBStd	4.34	2.43	2.68	3.11	2.72	2.64	2.51	3.17	3.02	2.36	2.77	3.09	2.31
	HDD10.0	1	1	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	394	122	83	44	20	5	0	0	0	0	2	32	87
	CDD10.0	3610	140	158	247	266	323	375	437	456	396	376	258	177
	CDD18.3	962	3	8	33	36	69	125	179	198	146	119	40	6
	CDH23.3	7858	59	128	429	370	527	869	1488	1647	991	908	371	72
CDH26.7	3079	6	26	170	159	198	298	651	741	336	363	124	7	
Wind	WSAvg	3.1	2.9	3.2	3.3	3.5	3.2	3.0	3.0	3.0	3.0	2.9	2.9	
Precipitation	PrecAvg	264	39	29	37	18	8	1	0	3	5	24	42	51
	PrecMax	559	170	169	171	121	118	12	1	27	98	131	359	169
	PrecMin	80	0	0	0	0	0	0	0	0	0	0	0	1
	PrecStd	144	45	40	42	28	24	2	0	7	20	33	71	49
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	27.3	29.8	34.0	36.4	38.9	40.0	45.5	43.8	38.6	37.0	33.1	27.4
		MCWB	13.4	13.9	15.7	16.7	18.0	19.6	20.3	20.7	20.7	17.8	16.1	14.4
	2%	DB	25.0	27.0	30.9	31.8	31.9	33.2	38.3	37.2	33.0	33.8	30.1	25.2
		MCWB	13.2	13.8	15.0	16.3	18.5	20.0	20.6	21.4	20.3	17.8	16.2	13.7
	5%	DB	22.9	24.7	28.2	26.9	27.2	29.1	32.1	32.9	29.8	30.9	27.8	23.4
		MCWB	12.6	13.8	14.9	16.5	18.5	19.9	20.7	21.3	20.2	17.5	15.7	13.6
	10%	DB	20.9	22.1	25.1	24.2	25.4	27.2	29.1	30.0	27.9	27.9	25.0	21.5
		MCWB	12.9	13.8	15.1	16.7	17.9	19.7	20.7	21.1	20.4	18.4	15.7	13.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	16.9	18.1	18.9	19.6	20.7	22.0	22.9	23.5	22.8	21.7	20.9	18.5
		MCDB	20.8	23.4	24.9	27.3	27.8	30.5	32.7	32.0	30.4	27.5	26.2	20.8
	2%	WB	15.6	16.8	17.7	18.5	19.7	21.2	22.1	22.6	21.7	20.9	19.3	17.3
		MCDB	19.9	21.0	23.8	25.2	26.8	28.6	30.6	30.8	28.0	26.4	23.5	20.3
	5%	WB	14.8	15.7	17.0	17.8	19.1	20.6	21.5	22.0	21.2	20.2	18.1	16.3
		MCDB	19.1	20.2	23.1	24.0	25.9	27.5	29.4	29.7	27.2	25.2	22.4	19.5
	10%	WB	13.9	15.0	16.2	17.2	18.5	20.0	21.1	21.5	20.7	19.6	17.2	15.3
		MCDB	18.4	19.6	21.8	23.0	24.7	26.4	28.4	28.6	26.3	24.4	21.9	18.9
Mean Daily Temperature Range	5% DB	MDBR	13.8	13.4	13.5	12.0	11.4	10.8	11.4	11.5	11.1	12.3	13.1	13.1
		MCDBR	16.3	17.0	18.1	16.5	14.7	14.3	16.5	15.7	14.0	16.6	15.9	15.6
		MCWBR	7.4	7.3	6.7	6.1	5.4	4.9	4.7	4.3	4.5	5.5	6.2	7.0
	5% WB	MCDBR	13.0	12.7	13.4	12.7	12.2	11.8	12.1	12.2	11.3	11.7	12.0	10.5
		MCWBR	7.3	6.9	6.7	5.9	5.1	4.5	4.4	4.4	4.4	5.1	6.2	5.8
Clear-Sky Solar Irradiance	taub	0.317	0.363	0.419	0.420	0.444	0.486	0.577	0.557	0.467	0.425	0.370	0.343	
	taud	2.512	2.355	2.161	2.194	2.139	2.030	1.798	1.869	2.110	2.221	2.376	2.451	
	Ebn at Noon	918	897	863	875	851	812	738	748	807	817	844	865	
	Edh at Noon	88	113	146	147	156	173	218	201	152	127	101	90	
All-Sky Solar Radiation	RadAvg	3.81	4.66	5.91	6.85	7.46	7.65	7.53	6.87	5.93	4.75	3.87	3.39	
	RadStd	0.21	0.36	0.24	0.27	0.37	0.17	0.19	0.17	0.28	0.21	0.23	0.25	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (3 neighbors)	+0.24	N/A	N/A	N/A	N/A	-0.32	N/A	N/A	+153	+171

Nomenclature: See separate page